Chapter 3

Stochastic Processes and Noise

3. Basic Concepts of Probability

Probability is a number that reflects the chance or likelihood that a particular event will occur. Probabilities can be expressed as proportions that range from 0 to 1, and they can also be expressed as percentages ranging from 0% to 100%. A probability of 0 indicates that there is no chance that a particular event will occur, whereas a probability of 1 indicates that an event is certain to occur. A probability of 0.45 (45%) indicates that there are 45 chances out of 100 of the event occurring.

Laplace's Classical Definition: The Probability of an event A is defined a-priori without actual experimentation as

$$P(A) = \frac{\text{Number of outcomes favorable to } A}{\text{Total number of possible outcomes}},$$

Provided all these outcomes are equally likely.

Relative Frequency Definition: The probability of an event A is defined as

$$P(A) = \lim_{n \to \infty} \frac{n_A}{n}$$

Where nA is the number of occurrences of A and n is the total number of trials.

Axioms of Probability

For any event A, we assign a number P(A), called the probability of the event A. This number satisfies the following three conditions that act the axioms of probability.

(i) $P(A) \geq 0$ (Probability is a nonnegative number)
(ii) $P(\Omega) = 1$ (Probability of the whole set is unity)
(iii) If $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$.

3.1 Sample Spaces, Events, and Their Probabilities

Rolling an ordinary six-sided die is a familiar example of a random experiment, an action for which all possible outcomes can be listed, but for which the actual outcome on any given trial of the experiment cannot be predicted with certainty. In such a situation we wish to assign to each

outcome, such as rolling a two, a number, called the probability of the outcome that indicates how likely it is that the outcome will occur. Similarly, we would like to assign a probability to any event, or collection of outcomes, such as rolling an even number, which indicates how likely it is that the event will occur if the experiment is performed. This section provides a framework for discussing probability problems, using the terms just mentioned.

Definition sample space

A random experiment is a mechanism that produces a definite outcome that cannot be predicted with certainty. The sample space associated with a random experiment is the set of all possible outcomes. An event is a subset of the sample space.

Definition of event

An event E is said to occur on a particular trial of the experiment if the outcome observed is an element of the set E.

Example 1

Construct a sample space for the experiment that consists of tossing a single coin.

Solution: The outcomes could be labeled h for heads and t for tails. Then the sample space is the set S={h,t}.

Example 2

Construct a sample space for the experiment that consists of rolling a single die. Find the events that correspond to the phrases “an even number is rolled” and “a number greater than two is rolled.”

Solution:

The outcomes could be labeled according to the number of dots on the top face of the die. Then the sample space is the set S={1,2,3,4,5,6}.

The outcomes that are even are 2, 4, and 6, so the event that corresponds to the phrase “an even number is rolled” is the set {2,4,6}, which it is natural to denote by the letter E. We write E={2,4,6}.

Similarly the event that corresponds to the phrase “a number greater than two is rolled” is the set T={3,4,5,6}, which we have denoted T.

3.2 Conditional Probability

Understanding Conditional Probability

Conditional probabilities are contingent on a previous result or event occurring. A conditional probability would look at such events in relationship with one another. Conditional probability is thus the likelihood of an event or outcome occurring *based on* the occurrence of some other event or prior outcome.

Dependent and Independent Events

The definition of independent and dependent events is connected to conditional probability. Let us see the definitions of independent and dependent events along with their formulas.

Dependent Events

Dependent events, as the name suggests, are any two events in which the happening of one event depends on the happening of the other event.

- If A depends on B, then the probability of A is P(A | B).
- If B depends on A, then the probability of B is P(B | A).

By the conditional probability formulas,

$$P(A \mid B) = P(A \cap B) / P(B) \Rightarrow P(A \cap B) = P(A \mid B) \cdot P(B)$$

$$P(B \mid A) = P(A \cap B) / P(A) \Rightarrow P(A \cap B) = P(B \mid A) \cdot P(A)$$

Thus, two event A and B are said to be dependent events if one of the conditions is satisfied.

- $P(A \cap B) = P(A \mid B) \cdot P(B)$ (or)
- $P(A \cap B) = P(B \mid A) \cdot P(A)$

Independent Events

Independent events, as the name suggests, are any two events in which the happening of one event does not depend on the happening of the other event. i.e., if A and B are independent then P(A | B) = P(A) and P(B | A) = P(B).

Thus, to get the formula of independent events, we just need to replace P(A | B) with P(A) (or P(B | A) with P(B)) in one of the above (dependent events) formulas. Hence, two events are said to be independent if

$$P(A \cap B) = P(A) \cdot P(B)$$

This is also called as multiplication rule of probability.

Conditional probability is often portrayed as the "probability of A *given* B," notated as P(A|B).

Suppose *A* and *B* are independent, then

Thus if *A* and *B* are independent, the event that *B* has occurred does not shed any more light into the event *A*. It makes no difference to *A* whether *B* has occurred or not.

$$P(A \mid B) = \frac{P(AB)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

An example will clarify the situation:

Example 5.2: A box contains 6 white and 4 black balls. Remove two balls at random without replacement. What is the probability that the first one is white and the second one is black?

Let W_1 = "first ball removed is white"

B_2 = "second ball removed is black"

Using the conditional probability rule,

$$P(W_1 B_2) = P(B_2 W_1) = P(B_2 \mid W_1) P(W_1)$$

But

$$P(W_1) = \frac{6}{6+4} = \frac{6}{10} = \frac{3}{5},$$

and

$$P(B_2 \mid W_1) = \frac{4}{5+4} = \frac{4}{9},$$

and hence

$$P(W_1 B_2) = \frac{3}{5} * \frac{4}{9} \approx 0.27$$

Are the event w1 and B2 are independent?

$$P(B_2) = P(B_2 \mid W_1) P(W_1) + P(B_2 \mid B_1) P(B_1)$$
$$= \frac{4}{5+4} \cdot \frac{3}{5} + \frac{3}{6+3} \cdot \frac{4}{10} = \frac{4}{9} \cdot \frac{3}{5} + \frac{1}{3} \cdot \frac{2}{5} = \frac{4+2}{15} = \frac{2}{5},$$

and

$$P(B_2)P(W_1) = \frac{2}{5} * \frac{3}{5} \neq P(B_2 W_1) = \frac{4}{15}$$

Since P(B2)P(W1)≠P(B2W1) the two event are dependent.

Conditional Probability Formula

$$P(B|A) = P(A \text{ and } B) / P(A)$$

Or:

$$P(B|A) = P(A \cap B) / P(A)$$

Where

P = *Probability*

A = *Event A*

B = *Event B*

Unconditional probability is also known as marginal probability and measures the chance of an occurrence ignoring any knowledge gained from previous or external events. Since this probability ignores new information, it remains constant.

Similarly, from

$$P(B \mid A) = \frac{P(BA)}{P(A)} = \frac{P(AB)}{P(A)},$$

or

$$P(AB) = P(B \mid A)P(A).$$

From, we get

$$P(A \mid B)P(B) = P(B \mid A)P(A)$$

or

$$P(A \mid B) = \frac{P(B \mid A)}{P(B)} \cdot P(A)$$

This is known as Bayes' theorem.

Although simple enough, Bayes' theorem has an interesting interpretation: P(A) represents the a-priori probability of the event A. Suppose B has occurred, and assume that A and B are not independent. How can this new information be used to update our knowledge about A? Bayes' rule in take into account the new information ("B has occurred") and gives out the a-posteriori probability of A given B. We can also view the event B as new knowledge obtained from a fresh experiment. We know something about A as P(A). The new information is available in terms of

B. The new information should be used to improve our knowledge/understanding of A. Bayes' theorem gives the exact mechanism for incorporating such new information.

KEY TAKEAWAYS

- Conditional probability refers to the chances that some outcome occurs given that another event has also occurred.
- It is often stated as the probability of B given A and is written as P(B|A), where the probability of B depends on that of A happening.
- Conditional probability can be contrasted with unconditional probability.
- Probabilities are classified as either conditional, marginal, or joint.

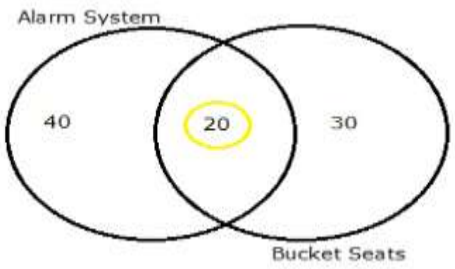
Example: In a group of 100 sports car buyers, 40 bought alarm systems, 30 purchased bucket seats, and 20 purchased an alarm system and bucket seats. If a car buyer chosen at random bought an alarm system, what is the probability they also bought bucket seats?

Step 1: Figure out P(A). It's given in the question as 40%, or 0.4.

Step 2: Figure out P(A∩B). This is the intersection of A and B: both happening together. It's given in the question 20 out of 100 buyers, or 0.2.

Step 3: Insert your answers into the formula:

$$P(B|A) = P(A \cap B) / P(A) = 0.2 / 0.4 = 0.5.$$



The probability that a buyer bought bucket seats, given that they purchased an alarm system, is 50%.

Examples of Conditional Probability

There are 12 balls in the pool. 6 balls are blue and the rest are red. Two of the balls are picked up from the pool without replacement. What is the probability of selecting two red balls from the pool?

Solution

We will calculate the probability of the first ball.

Total number of balls in the pool = 12

Probability that a randomly selected ball is red = $\frac{6}{12} = \frac{1}{2}$

Now, we will compute the probability of the second ball.

Total number of balls left in the pool = 11

Number of red balls left in the pool = 5

Probability of selecting a red ball second time = $\frac{5}{11}$

Probability that two balls selected randomly are red in color = P(A) x P(B|A) = $\frac{1}{2} \cdot \frac{5}{11} = \frac{5}{22}$

Example

In a class, 35% of the students study science and history. 65% of the students study science. What is the probability of a student studying history given he/she is already studying science?

Solution

Percentage of the students studying science and history = 35%

Probability of students studying science and history = P(S and H) = 0.35

Percentage of students studying science = 65%

The probability of students studying science = P(S) = 0.65

$$P(H|S) = \frac{P(S \quad and \quad H)}{P(S)} = \frac{0.35}{0.65} = 0.53$$

Random variable

What Is a Random Variable?

A random variable is a variable whose value is unknown or a function that assigns values to each of an experiment's outcomes.

Type of random variable

Random variables are often designated by letters and can be classified as **discrete**, which are variables that have specific values, or **continuous**, which are variables that can have any values within a continuous range.

EXAMPLE 2.1 Suppose that a coin is tossed twice so that the sample space is S = {HH, HT, TH, TT}. Let X represent the number of heads that can come up. With each sample point we can associate a number for X as shown in Table 2-1. Thus, for example, in the case of HH (i.e., 2 heads), X = 2 while for TH (1 head), X = 1. It follows that X is a random variable.

Sample Point	*HH*	*HT*	*TH*	*TT*
X	2	1	1	0

It should be noted that many other random variables could also be defined on this sample space, for example, the square of the number of heads or the number of heads minus the number of tails. A random variable that takes on a finite or countable infinite number of values is called a discrete random variable while one which takes on a non-countable infinite number of values is called a non-discrete random variable.

KEY TAKEAWAYS

- A random variable is a variable whose value is unknown or a function that assigns values to each of an experiment's outcomes.
- A random variable can be either discrete (having specific values) or continuous (any value in a continuous range).
- The use of random variables is most common in probability and statistics, where they are used to quantify outcomes of random occurrences.
- Risk analysts use random variables to estimate the probability of an adverse event occurring.

Understanding a Random Variable

In probability and statistics, random variables are used to quantify outcomes of a random occurrence, and therefore, can take on many values. Random variables are required to be measurable and are typically real numbers. For example, the letter X may be designated to represent the sum of the resulting numbers after three dice are rolled.

In this case, X could be 3 (1 + 1+ 1), 18 (6 + 6 + 6), or somewhere between 3 and 18, since the highest number of a die is 6 and the lowest number is 1.

Types of Random Variables

Discrete Random Variables

Discrete random variables take on a countable number of distinct values. Consider an experiment where a coin is tossed three times. If X represents the number of times that the coin comes up heads, then X is a discrete random variable that can only have the values 0, 1, 2, or 3 (from no heads in three successive coin tosses to all heads). No other value is possible for X.

Continuous Random Variables

Continuous random variables can represent any value within a specified range or interval and can take on an infinite number of possible values. An example of a continuous random variable would be an experiment that involves measuring the amount of rainfall in a city over a year or the average height of a random group of 25 people.

Drawing on the latter, if Y represents the random variable for the average height of a random group of 25 people, you will find that the resulting outcome is a continuous figure since height may be 5 ft or 5.01 ft or 5.0001 ft. Clearly, there is an infinite number of possible values for height.

Example of a Random Variable

A typical example of a random variable is the outcome of a coin toss. Consider a probability distribution in which the outcomes of a random event are not equally likely to happen. If the random variable Y is the number of heads we get from tossing two coins, then Y could be 0, 1, or 2. This means that we could have no heads, one head, or both heads on a two-coin toss.

However, the two coins land in four different ways: TT, HT, TH, and HH. Therefore, the P(Y=0) = 1/4 since we have one chance of getting no heads (i.e., two tails [TT] when the coins are tossed). Similarly, the probability of getting two heads (HH) is also 1/4. Notice that getting one head has a likelihood of occurring twice: in HT and TH. In this case, P (Y=1) = 2/4 = 1/2.

What Is a Mixed Random Variable?

A mixed random variable combines elements of both discrete and continuous random variables.

How Do You Identify a Random Variable?

A random variable is one whose value is unknown a priori, or else is assigned a random value based on some data generating process or mathematical function.

Why Are Random Variables Important?

Random variables produce probability distributions based on experimentation, observation, or some other data-generating process. Random variables, in this way, allow us to understand the world around us based on a sample of data, by knowing the likelihood that a specific value will occur in the real world or at some point in the future.

3.2.1 Probability distribution

Probability distribution is an assignment of probabilities to the values of the random variable. The abbreviation of pdf is used for a probability distribution function.

For probability distributions, $0 \leq P(x) \leq 1$ and $\sum P(x) = 1$

Example: A fair coin is tossed twice, and let the r.v *X* represent the number of heads. Find $F_x(X)$

Solution: In this case $\Omega = \{HH, HT, TH, TT\}$, and

$$X(HH) = 2, X(HT) = 1, X(TH) = 1, X(TT) = 0.$$

$$x < 0, \quad \{X(\xi) \leq x\} = \phi \Rightarrow F_X(x) = 0,$$

$$0 \leq x < 1, \{X(\xi) \leq x\} = \{TT\} \Rightarrow F_X(x) = P\{TT\} = P(T)P(T) = \frac{1}{4},$$

$$1 \leq x < 2, \{X(\xi) \leq x\} = \{TT, HT, TH\} \Rightarrow F_X(x) = P\{TT, HT, TH\} = \frac{3}{4},$$

$$x \geq 2, \quad \{X(\xi) \leq x\} = \Omega \Rightarrow F_X(x) = 1. \text{ (Fig. 3.4)}$$

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$$P\{X = 1\} = F_X(1) - F_X(1^-) = 3/4 - 1/4 = 1/2.$$

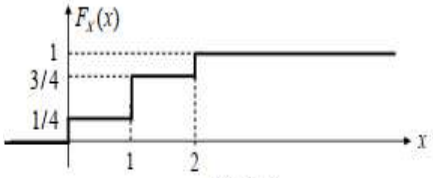


Fig. 6.4

Probability density function (p.d.f)

The derivative of the distribution function is called the probability density function of the r.v X. Thus

$$f_X(x) \triangleq \frac{dF_X(x)}{dx}$$

3.3 Function of random variable

In this Section, we study further properties of expectations of random variables. We move on from the expectation of a single random variable to consider the expectation of the function of a collection of random variables, X1,X2,…,Xn. We pay particular attention to the expectation of functions of two random variables X and Y, say. We define the **covariance** as a measure of how the **random variables X and Y vary together** and the correlation which provides a measure of linear **dependence between two random variables X and Y**.

Expectation of a function of random variables

The expectation (mean) of X is defined as,

$$\mathrm{E}(X) = \frac{\sum_{i=1}^{n} X_i}{n}$$

For a continuous random variable X having density function f(x), the expectation of X is

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$$

Provided that the integral converges absolutely.

The expectation of X is very often called the mean of X and is denoted by X, or simply, when the particular random variable is understood. The mean, or expectation, of X gives a single value that acts as a representative or average of the values of X, and for this reason it is often called a measure of central tendency.

Rule 1: The expectation of a constant, c, is the constant.

$$\mathbb{E}(c) = c$$

Rule 2: Adding a constant value, c, to each term in a random variable, X, increases the expected value (or mean) of X by the constant.

$$\mathbb{E}(X + c) = \mathbb{E}(X) + c$$

Rule 3: Multiplying a random variable, X, by a constant value, c, multiplies the expected value (or mean) of X by that constant.

$$\mathbb{E}(cX) = c(\mathbb{E}(X))$$

Rule 4: The expected value (or mean) of the sum of two random variables, X and Y is the sum of the expected values (or means). This is also known as the *additive law of expectation*.

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$$

Example: The density function of a random variable X is given by

$$f(x) = \begin{cases} \frac{1}{2}x & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

The expected value of X is then

$$E(X) = \int_{-\infty}^{\infty} xf(x)\,dx = \int_0^2 x\left(\frac{1}{2}x\right)dx = \int_0^2 \frac{x^2}{2}\,dx = \frac{x^3}{6}\Big|_0^2 = \frac{4}{3}$$

Example:

The joint density function of two random variables X and Y is given by

$$f(x, y) = \begin{cases} xy/96 & 0 < x < 4, 1 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

Find (a) $E(X)$, (b) $E(Y)$, (c) $E(XY)$, (d) $E(2X + 3Y)$.

(a) $$E(X) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x, y)\,dx\,dy = \int_{x=0}^{4}\int_{y=1}^{5} x\left(\frac{xy}{96}\right)dx\,dy = \frac{8}{3}$$

(b) $$E(Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf(x, y)\,dx\,dy = \int_{x=0}^{4}\int_{y=1}^{5} y\left(\frac{xy}{96}\right)dx\,dy = \frac{31}{9}$$

Covariance

Another useful measure that we will be working with in the course is the covariance. We denote the covariance between X and Y using σXY or Cov(X, Y).

The most common formula for covariance is:

$$\sigma_{XY} = \mathrm{Cov}(X,Y) = \frac{\sum_{i=1}^{n}\left(X_i - \mathbb{E}(X)\right)\left(Y_i - \mathbb{E}(Y)\right)}{n}$$

The covariance can also be expressed as an expectation:

$$\sigma XY = \mathrm{Cov}(X, Y) = \mathbb{E}([X - \mathbb{E}(X)][Y - \mathbb{E}(Y)])$$

Lastly, we can also express the covariance as a difference of expectations.

$$\sigma XY = \mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

Rule 1: The covariance of two constants, c and k, is zero.

$$\mathrm{Cov}(c, k) = 0$$

Rule 2: The covariance of two independent random variables is zero.

$$\mathrm{Cov}(X, Y) = 0$$

Rule 3: The covariance is a combinative.

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$$

Rule 4: The covariance of a random variable, X, with a constant, c is zero.

$$\mathrm{Cov}(X, c) = 0$$

Rule 5: Adding a constant to either or both random variables does not change their covariances.

$$\mathrm{Cov}(X + c, Y + k) = \mathrm{Cov}(X, Y)$$

Rule 6: Multiplying a random variable by a constant multiplies the covariance by that constant.

$$\mathrm{Cov}(cX, kY) = c \times k \times \mathrm{Cov}(X, Y)$$

Rule 7: The additive law of covariance holds that the covariance of a random variable with a sum of random variables is just the sum of the covariances with each of the random variables.

$$\mathrm{Cov}(X + Y, Z) = \mathrm{Cov}(X, Z) + \mathrm{Cov}(Y, Z)$$

Rule 8: The covariance of a variable with itself is the variance of the random variable.

$$\mathrm{Cov}(X, X) = \mathrm{Var}(X)$$

Correlation

The correlation coefficient is a standardized covariance value. We denote the correlation between X and Y using ρXY or Cor(X, Y). The most common formula for correlation is:

$$\rho(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}$$

Rules for Working with Correlation Coefficients

Rule 1: Adding a constant to a random variable does not change their correlation coefficient.

$$\text{Cor}(X + c, Y + k) = \text{Cor}(X, Y)$$

Rule 2: Multiplying a random variable by a constant does not change their correlation coefficient.

$$\text{Cor}(cX, dY) = \text{Cor}(X, Y)$$

Rule 3: Because the square root of the variance is always positive, the correlation coefficient can be negative only when the covariance is negative. This implies that:

$$-1 \le \text{Cor}(X, Y) \le 1$$

Example

Suppose that the random variables X and Y have a joint density function given by

$$f(x, y) = \begin{cases} c(2x + y) & 2 < x < 6, 0 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

Using c = 1 / 210, we have:

Find (a) $E(X)$, (b) $E(Y)$, (c) $E(XY)$, (d) $E(X^2)$, (e) $E(Y^2)$, (f) Var (X), (g) Var (Y), (h) Cov (X, Y), (i) ρ,

Solution

(a) $$E(X) = \frac{1}{210}\int_{x=2}^{6}\int_{y=0}^{5}(x)(2x + y)\,dx\,dy = \frac{268}{63}$$

(b) $$E(Y) = \frac{1}{210}\int_{x=2}^{6}\int_{y=0}^{5}(y)(2x + y)\,dx\,dy = \frac{170}{63}$$

(c) $$E(XY) = \frac{1}{210}\int_{x=2}^{6}\int_{y=0}^{5}(xy)(2x + y)\,dx\,dy = \frac{80}{7}$$

(d) $$E(X^2) = \frac{1}{210}\int_{x=2}^{6}\int_{y=0}^{5}(x^2)(2x + y)\,dx\,dy = \frac{1220}{63}$$

(e) $$E(Y^2) = \frac{1}{210}\int_{x=2}^{6}\int_{y=0}^{5}(y^2)(2x + y)\,dx\,dy = \frac{1175}{126}$$

(f) $$\sigma_X^2 = \text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1220}{63} - \left(\frac{268}{63}\right)^2 = \frac{5036}{3969}$$

(g) $$\sigma_Y^2 = \text{Var}(Y) = E(Y^2) - [E(Y)]^2 = \frac{1175}{126} - \left(\frac{170}{63}\right)^2 = \frac{16,225}{7938}$$

(h) $$\sigma_{XY} = \text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{80}{7} - \left(\frac{268}{63}\right)\left(\frac{170}{63}\right) = -\frac{200}{3969}$$

(i) $$\rho = \frac{\sigma_{XY}}{\sigma_X\sigma_Y} = \frac{-200/3969}{\sqrt{5036/3969}\sqrt{16,225/7938}} = \frac{-200}{\sqrt{2518}\sqrt{16,225}} = -0.03129 \text{ approx.}$$

Stochastic Process Definition (reading assignment)

A stochastic process, also known as a random process, is a collection of random variables that are indexed by some mathematical set. Each probability and random process are uniquely associated with an element in the set. The index set is the set used to index the random variables. The index set was traditionally a subset of the real line, such as the natural numbers, which provided the index set with time interpretation. Stochastic Process Meaning is one that has a system for which there are observations at certain times, and that the outcome, that is, the observed value at each time is a random variable. Each random variable in the collection of the values is taken from the same mathematical space, known as the state space.

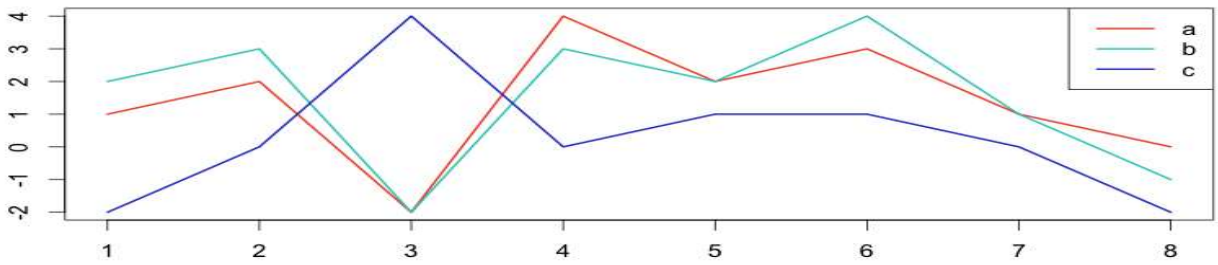
Auto correlation and cross correlation

Definition:

Cross-correlation is the comparison of two different time series to detect if there is a correlation between metrics with the same maximum and minimum values.

To detect a level of correlation between two signals we use cross-correlation. It is calculated simply by multiplying and summing two-time series together.

In the following example, graphs A and B are cross-correlated but graph C is not correlated to either. For example: “Are two audio signals in phase?”



This is a kind of correlation, in which the signal in-hand is correlated with another signal so as to know how much resemblance exists between them. Mathematical expression for the cross-correlation of continuous time signals x (t) and y (t) is given by

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t)\, y^{\star}(t-\tau)\, dt$$

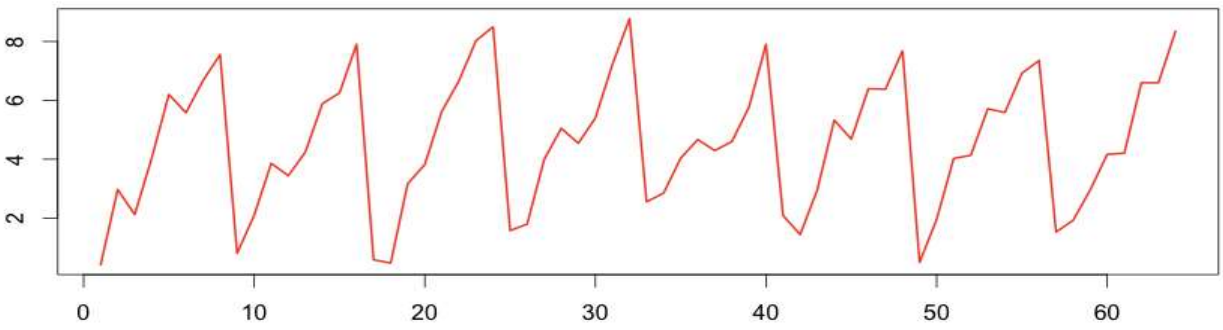
Similarly, the cross-correlation of the discrete time signals x [n] and y [n] is expressed as

$$R_{xy}[m] = \sum_{n=-\infty}^{\infty} x[n]\, y^{\star}[n-m]$$

Auto-correlation is the comparison of a time series with itself at a different time. It aims, for example, to detect repeating patterns or seasonality.

Auto-correlation is very useful in many applications; a common one is detecting repeatable patterns due to seasonality. The following graph clearly shows repeating patterns every 8 data points. Indeed, looking at the R code, it's a repeatable sequence of the numbers 1 through 8 with some random noise in the mix.

For example: "Is there weekly seasonality on a server website?" "Does the current week's data highly correlate with that of the previous week?"



This is a type of correlation in which the given signal is correlated with itself, usually the time-shifted version of itself. Mathematical expression for the autocorrelation of continuous time signal x (t) is given by

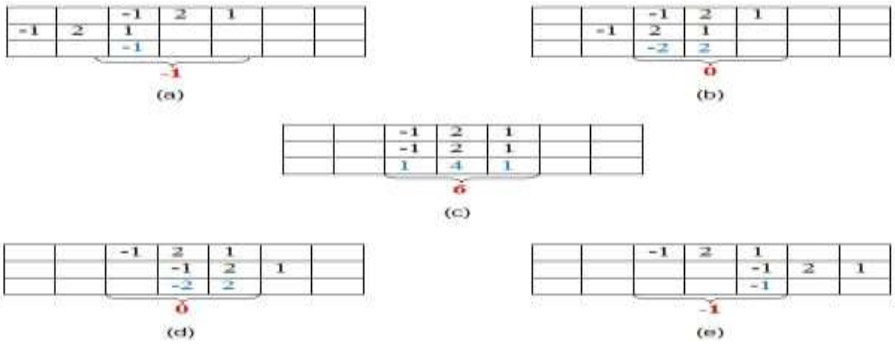
$$R_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)\, x^{\star}(t-\tau)\, dt$$

Where $\star$ denotes the complex conjugate.

Similarly the autocorrelation of the discrete time signal x[n] is expressed as

$$R_{xx}[m] = \sum_{n=-\infty}^{\infty} x[n]\, x^{\star}[n-m]$$

For example, autocorrelation of the digital signal x [n] = {-1, 2, 1} can be computed as shown in Figure 1.



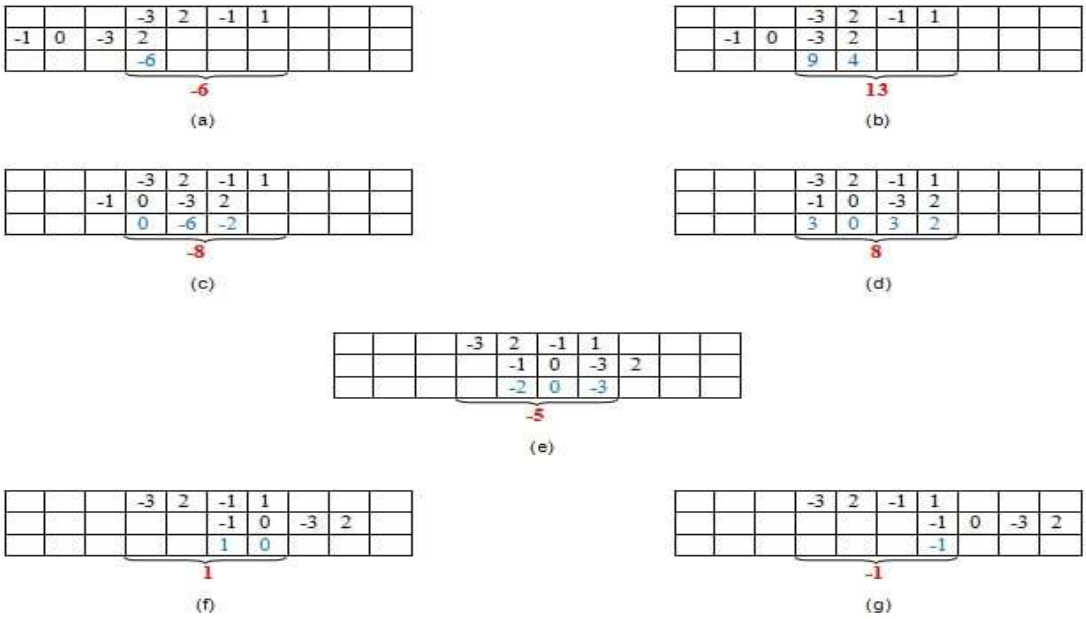
Here, the first set of samples (those in the first row of every table) refers to the given signal. The second set (in the second row of every table) refers to the samples of its time-shifted version. Next, the samples shown in red color in the third row are obtained by multiplying the corresponding samples of the first two rows.

Finally, we add the samples in the last row of the sample (contained within the curly brackets) so as to obtain the samples of the auto-correlated signal.

Thus, here we find that the samples of the autocorrelated signal Rxx are {-1, 0, 6, 0, -1}, where 6 is the zeroth sample.

The example presented shows that the sample of the autocorrelated signal will be at its maximum value when the overlapping signal best matches the given signal. In this case, it happens when time-shift is zero.

For example, cross-correlation of the digital signals x [n] = {-3, 2, -1, 1} and y [n] = {-1, 0, -3, 2} can be computed as shown by Figure 2.



Here, the first set of samples (in the first row of every table) refers to the signal x [n] and the second set refers to the samples (in the second row of every table) of the signal y [n].

Next, the samples shown in blue color—those in the third row—are obtained by multiplying the corresponding samples of the first two rows. Finally, we add the samples in the last row (contained within the curly brackets) so as to obtain the samples of the cross-correlated signal.

Thus, here we see that the samples of the cross-correlated signal Rxy are obtained as {-6, 13, -8, 8, -5, 1, -1}, where 8 is the zeroth sample. Further, the example presented shows that the sample of the cross-correlated signal is at its highest peak, with value 13, when the last two samples of y [n] overlap with the first two samples of x [n]. This is because, in this case, the second signal overlaps with the first at its best, as the two samples in each of the signals are identical. Hence, it can be concluded that the cross-correlation reaches its maximum when the two signals considered become most similar to each other.

Assignments content

1, Stationary Processes

2, Power spectral density

3, Systems with Random Signal Excitation

4, Electrical Noise

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There will be presentation and the presenter will be randomly selected.